# JAMES SUD

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## EDUCATION

**University of Chicago**: Ph.D. student in Computer Science. Advisor: Fred Chong **2022-Present**

**University of Chicago:** Masters in Computer Science. **2025**

**University of California, Berkeley**: B.S. in Physics, Minor in Computer Science. GPA: 3.9. **2020**

## RESEARCH EXPERIENCE

Infleqtion **June 2026 – present**

Computer Science Department, University of Chicago (NSF GRFP fellow) **Aug 2022 – present**

JP Morgan Chase Global Technology Applied Research **Summer 2024 and 2025**

Universities Space Research Association, NASA Academic Mission Services **May 2020 – Aug 2022**

Los Alamos National Laboratory Quantum Computing Summer School **Jun 2020 – Aug 2020**

Lawrence Berkeley National Laboratory **Aug 2017 – June 2020**

## PUBLICATIONS/PREPRINTS

Authors ordered by contribution except (a) denoting alphabetical. Bold: first/corresponding author.

1. Kunal Marwaha, **JS** (a). "A complexity phase transition at the EPR Hamiltonian." 2026. arXiv: 2604.13026.
2. Anuj Apte, Eunou Lee, Ojas Parekh, Kunal Marwaha, Lennart Singorjo, **JS** (a)**.** "A 0.8395-approximation algorithm for the EPR problem." 2026. arXiv:2512.09896
3. Anuj Apte, Ojas Parekh, **JS** (a). "Conjectured Bounds for 2-Local Hamiltonians via Token Graphs." 2025. arXiv:2506.03441.
4. Anuj Apte, Eunou Lee, Ojas Parekh, Kunal Marwaha, **JS** (a)**.** "Improved Algorithms for Quantum MaxCut via Partially Entangled Matchings." 2025. European Symposium on Algorithms.
5. Anuj Apte, et. al. "Iterative Interpolation Schedules for Quantum Approximate Optimization Algorithm." 2025. ACM Transactions on Quantum Computing. arXiv:2504.01694
6. Sami Boulebnane, JS, Ruslan Shaydulin, Marco Pistoia. "Equivalence of Quantum Approximate Optimization Algorithm and Linear-Time Quantum Annealing for the Sherrington-Kirkpatrick Model." 2025. arXiv:2503.09563.
7. Mariesa H Teo, Willers Yang, James Sud, Teague Tomesh, Frederic T Chong, Eric R Anschuetz. "k-Contextuality as a Heuristic for Memory Separations in Learning." 2025. 2025 IEEE International Conference on Quantum Computing and Engineering.

8. Kunal Marwaha, Adrian She, **JS** (a). "Performance of Variational Algorithms for Local Hamiltonian Problems on Random Regular Graphs." 2024. arXiv:2412.15147.
9. Daniel Belkin et. al. "Approximate t-Designs in Generic Circuit Architectures." 2024. PRX Quantum.
10. Filip B Maciejewski et. al. "Design and execution of quantum circuits using tens of superconducting qubits and thousands of gates for dense Ising optimization problems." 2024. Physical Review Applied.
11. Eric Anschuetz et. al. "Quantifying the Limits of Classical Machine Learning Models Using Contextuality." 2024. IEEE International Conference on Quantum Computing and Engineering (QCE)
12. Srikar Kasi et. al. "A Quantum Approximate Optimization Algorithm-based Decoder Architecture for NextG Wireless Channel Codes." 2024. IEEE International Conference on Quantum Computing and Engineering.
13. Bram Evert et. al. "Syncopated dynamical decoupling for suppressing crosstalk in quantum circuits." 2024. Physical Review Applied.
14. Siddharth Dangwal et. al. "COMPASS: Compiler Pass Selection For Improving Fidelity Of NISQ Applications." 2024. IEEE International Conference on Rebooting Computing (ICRC)
15. Yizhi Shen et. al. "Real-time Krylov theory for quantum computing algorithms." 2023. Quantum.
16. **JS** et. al. "A Parameter Setting Heuristic for the Quantum Alternating Operator Ansatz." 2023. Physical Review Research
17. Plato Deliyannis et. al. "Improving Quantum Simulation Efficiency of Final State Radiation with Dynamic Quantum Circuits." 2022. Physical Review D
18. **JS** et. al. "Dual-map framework for noise characterization of quantum computers." 2022. Physical Review A
19. Sohaib Alam et. al. "Practical verification of quantum properties in quantum-approximate-optimization runs." 2022. Physical Review Applied.
20. **JS** et. al. "Correlation-informed permutation of qubits for reducing ansatz depth in the variational quantum eigensolver." 2021. PRX Quantum.

## TALKS

1. "A complexity phase transition at the EPR Hamiltonian."
   a. Phasecraft. June 2026.
   b. University of Maryland, College Park. June 2026.
2. "Optimizing Local Hamiltonians." UChicago Computer Science. May 2026.
3. "Classical descriptions of Hamiltonian optimization." Korea Institute for Advanced Study. Jan 2026.
4. "Playing Checkers on Graphs." UChicago Computer Science. Oct 2025.
5. "Improved Algorithms for Quantum MaxCut via Partially Entangled Matchings." Sept 2025. European Symposium on Algorithms.
6. "Performance of Algorithms for Quantum Constraint Satisfaction Problems." UChicago Computer Science. April 2025.
7. "Analysis of the Quantum Approximate Optimization Algorithm with Constant Evolution Time." American Physical Society Global Physics Summit. Mar 2025.

8. “Average-case algorithms for quantum constraint-satisfaction problems.” UChicago Pritzker School of Molecular Engineering. Dec 2024.
9. “Towards Classical Descriptions of the Quantum Approximate Optimization Algorithm” American Physical Society Global Physics Summit. Mar 2024.
10. "Frustration." UChicago Computer Science. Feb 2024.
11. “Estimating the Dynamics of the Quantum Approximate Optimization Algorithm.” UChicago Computer Science. Mar 2023.
12. “Towards Classical Descriptions of the Quantum Approximate Optimization Algorithm.” UChicago Pritzker School of Molecular Engineering. Nov 2023.
13. “A Parameter Setting Heuristic for the Quantum Alternating Operator Ansatz.” American Physical Society Global Physics Summit. Mar 2023.